

Appendix I (Handouts-3): Applications of Matrices and System of Linear Equations in Genetic Engineering and Biotechnology

Introduction

Matrices and systems of linear equations have significant applications in modern biotechnology, including DNA analysis, gene expression studies, protein synthesis, enzyme kinetics, metabolic pathway analysis, cell population modeling, drug formulation, and bioinformatics. The following problems are designed to demonstrate how mathematical tools can be applied to biological systems.

Section A: Systems of Linear Equations

Problem 1: DNA Nucleotide Composition Analysis

A biotechnology laboratory is analyzing three DNA samples. Each sample contains different amounts of three nucleotides: Adenine (A), Thymine (T), and Cytosine (C).

The compositions of the samples are shown below:

DNA Sample	Adenine (A)	Thymine (T)	Cytosine (C)	Total Measured Amount
Sample I	2	3	1	13
Sample II	1	2	3	14
Sample III	3	1	2	13

Form a system of linear equations and solve it using Gaussian elimination /Gauss-Jordan elimination/the inverse matrix method, or Cramer's rule to determine the amount of Adenine, Thymine, and Cytosine present in the DNA samples.

Problem 2: CRISPR Gene Editing Reagent Preparation

A researcher prepares three reagent mixtures X, Y and Z.

Mixture	Guide RNA	Cas enzyme	Buffer
X	2	1	3
Y	1	3	2
Z	3	2	1

A final CRISPR experiment requires 14 units of guide RNA, 14 units of Cas enzyme, and 14 units of buffer.

Determine the required amount of each reagent mixture.

Problem 3: Enzyme Production by Microbial Strains

Three microbial strains A, B and C produce three enzymes. The production rates (units per liter) are given below:

Strain	Enzyme 1	Enzyme 2	Enzyme 3
A	2	1	3
B	1	4	2
C	3	2	1

A biotechnology company needs 17 units of Enzyme 1, 16 units of Enzyme 2, and 13 units of Enzyme 3.

Determine the volume of each microbial strain required.

Section B: Matrix Operations (Exercises)

Problem 4: Gene Expression Matrix Analysis

The expression levels of three genes (G_1 , G_2 , G_3) under three environmental conditions (C_1 , C_2 , C_3) are represented by:

	G_1	G_2	G_3
C_1	12	8	5
C_2	6	15	7
C_3	9	4	11

Determine:

- Total expression of each gene
- Total expression under each condition
- Average expression of each gene
- The most highly expressed gene

Problem 5: Genetic Profile Transformation

A genetic profile matrix is given by:

$$A = \begin{bmatrix} 4 & 2 & 3 \\ 1 & 5 & 2 \\ 3 & 1 & 4 \end{bmatrix}$$

Matrix A

Rows $\rightarrow$ Samples

Columns $\rightarrow$ Genes

Entries $\rightarrow$ Gene expression levels

A treatment effect matrix is:

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Matrix B

Rows $\rightarrow$ Old genes

Columns $\rightarrow$ New genes after treatment

Entries $\rightarrow$ Influence coefficients

Find the transformed genetic profile AB .

Matrix AB

Rows $\rightarrow$ Samples

Columns $\rightarrow$ New gene expression levels after treatment

Problem 6: Cell Growth and Transformation Problem

In a biotechnology experiment, scientists are studying how a group of cells changes during growth.

There are **three types of cells**:

- Stem Cells (C_1)
- Intermediate Cells (C_2)
- Mature Cells (C_3)

At the beginning of the experiment, the number of cells is:

$$P_0 = \begin{bmatrix} 1000 \\ 800 \\ 500 \end{bmatrix}$$

This means:

- 1000 Stem Cells
- 800 Intermediate Cells
- 500 Mature Cells

How Cells Change

During one growth cycle, cells may change from one type to another. The following table shows how cells transform:

$$T = \begin{bmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.3 & 0.7 \end{bmatrix}$$

Each column describes what happens to one type of cell:

- **First column (Stem Cells):**
 - 60% stay as stem cells
 - 30% become intermediate cells
 - 10% become mature cells
- **Second column (Intermediate Cells):**

- 20% become stem cells
- 50% stay intermediate cells
- 30% become mature cells
- **Third column (Mature Cells):**
 - 10% become stem cells
 - 20% become intermediate cells
 - 70% stay mature cells

Task

After one growth cycle, find the new number of:

- Stem Cells
- Intermediate Cells
- Mature Cells

Hint

Use the matrix multiplication:

$$P_1 = TP_0$$

Section C: Determinants and Inverse Matrices

Problem 7: Protein Transformation Analysis

A protein interaction matrix is given by

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Find:

- a) $\det(A)$
- b) Determine whether the matrix is invertible
- c) Find A^{-1} using a suitable method

Problem 8: Nutrient Solution Preparation

A laboratory prepares a culture medium using three supplements.

Supplement	Amino acids	Vitamins	Minerals
A	3	2	1
B	2	4	3
C	1	3	5

A culture requires 14 units of amino acids, 19 units of vitamins, and 20 units of minerals.

Determine the amount of each supplement needed.

Section D: Rank and Linear Dependence

Problem 9: Genetic Data Redundancy

A genetic information matrix is

$$D = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}$$

Find the rank of the matrix and explain whether the data contain redundant genetic information.

Problem 10: Gene Interaction Network

The interaction between four genes and three regulatory factors is represented by

$$M = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 0 \\ 1 & 3 & 2 \\ 5 & 7 & 2 \end{bmatrix}$$

Determine the rank of the matrix and identify whether the regulatory information is independent.

Section E: Advanced Applications

Problem 11: DNA Mutation Model (Eigenvalues)

The mutation transition matrix is

$$M = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

Find the eigenvalues and eigenvectors and discuss the long-term mutation behavior.

Problem 12: Metabolic Pathway Analysis

The conversion of three metabolites A, B, C is represented by:

$$\begin{bmatrix} A' \\ B' \\ C' \end{bmatrix} = \begin{bmatrix} 0.5 & 0.2 & 0.1 \\ 0.3 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix}$$

If the initial concentrations are A=50, B=30, and C=20, determine the concentrations after one reaction cycle.

Section F: PCR, Gene Expression, and Protein Synthesis

Problem 13: PCR Amplification Analysis (Matrix Application)

A biotechnology laboratory monitors the number of DNA copies produced after three PCR cycles under three different temperature conditions. The data are represented by

$$P = \begin{bmatrix} 200 & 250 & 300 \\ 400 & 450 & 500 \\ 800 & 900 & 1000 \end{bmatrix}$$

Determine:

- The total DNA copies produced at each temperature condition.
- The average DNA amplification after each cycle.
- Identify the condition that gives the highest amplification.

Problem 14: mRNA and Protein Production (System of Linear Equations)

Three genes G_1 , G_2 , and G_3 contribute to the production of three proteins P_1 , P_2 , and P_3 .

The contribution matrix is:

$$2G_1 + G_2 + 3G_3 = 20$$

$$G_1 + 3G_2 + 2G_3 = 18$$

$$3G_1 + 2G_2 + G_3 = 19$$

Determine the expression levels of G_1 , G_2 , and G_3 .

Problem 15: Gene Expression Comparison Using Matrix Addition

Gene expression levels of three genes before and after a treatment are represented by:

$$A = \begin{bmatrix} 10 & 15 & 20 \\ 12 & 18 & 22 \\ 8 & 11 & 16 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

Find:

- $A+B$, representing the total expression level.
- $A-B$, representing the change in expression due to treatment.

Section G: Drug Formulation and Biomedical Engineering

Problem 16: Antibiotic Mixture Optimization (System of Linear Equations)

Three antibiotic solutions A, B and C contain different amounts of three active compounds X, Y and Z.

Solution	X	Y	Z
A	3	1	2
B	1	2	4
C	2	3	1

A treatment requires 18 units of X, 19 units of Y, and 16 units of Z.

Determine the amount of each antibiotic solution required.

Problem 17: DNA Sequence Similarity Matrix

The similarity scores among three DNA sequences are represented by:

$$S = \begin{bmatrix} 1 & 0.8 & 0.6 \\ 0.8 & 1 & 0.7 \\ 0.6 & 0.7 & 1 \end{bmatrix}$$

Find:

- The determinant of S.
- Comment on whether the similarity information is independent.

Problem 18: Genome Data Compression (Rank of Matrix)

A genome information matrix is:

$$G = \begin{bmatrix} 4 & 8 & 12 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

Determine:

- The rank of the matrix.
- Explain the significance of the result in terms of data redundancy.

Problem 19: Gene Regulatory Network

The interaction between four genes and three regulatory proteins is represented by

$$R = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 3 & 1 & 2 \\ 5 & 5 & 2 \end{bmatrix}$$

Find the rank of the matrix and discuss whether the regulatory signals are independent.

Problem 20: Population Transition Model

A population of three cell types changes according to

$$P_{n+1} = AP_n$$

where

$$A = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.2 & 0.7 \end{bmatrix}$$

If the initial population vector is

$$P_0 = \begin{bmatrix} 1000 \\ 600 \\ 400 \end{bmatrix},$$

determine the population after one cycle.

Problem 21: Mutation Probability Matrix (Eigenvalue Problem)

The mutation process is represented by:

$$M = \begin{bmatrix} 0.85 & 0.15 \\ 0.25 & 0.75 \end{bmatrix}$$

Determine the eigenvalues and eigenvectors of the mutation matrix and discuss their biological interpretation.